

Jake Hoffman

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LinkedIn

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SUMMARY OF QUALIFICATIONS

- Computer Engineering student with a strong foundation in software engineering principles, technical problem solving, and design analysis.
- Proficient in Python, Java, C, C#, C++, SQL, Git, Word, Excel, JavaScript, Dart, Unity, FastAPI, Playwright, Monday.com, and Google Apps Script.
- Writes clean and maintainable code with attention to testing, logging, and performance.

EDUCATION

3rd Year at Queen's University

September 2023 – Present

Computer Engineering

- Cumulative GPA of 3.85 with distinction of Dean's Scholar.
- Developed a pathfinding algorithm in Python for autonomous operation of a rover.

EXPERIENCE

Independent Software Engineer – Trading & Automation Systems

May 2026 – Present

Self-directed, full-time

- Operate two 24/7 production systems: trader, an S&P 500 paper-trading agent (750+ commits since April 2026), and a ten-bookmaker arbitrage daemon in production since November 2025 (3,900+ commits).
- Took market-bot (crypto market-making feasibility study) from first commit to shipped verdict in three weeks: 7 days of 24/7 venue telemetry, 9-criterion pre-trade synthesizer, data-backed NOT_READY verdict across all 75 pairs.

IT Intern

Jun 2024 – Aug 2024; Jun 2025 – Aug 2025

Lynx Equity Limited

- Developed a task management system on Monday.com, streamlining communication between subsidiary companies and internal departments by automating issue routing to the appropriate workflow queue.
- Built an On/Off-Boarding workflow in Monday.com, standardizing role-based tasks, handoffs, and approvals; delivered a stakeholder demo and incorporated feedback into the final product.
- Wrote Google Apps Script (JavaScript) that consolidates multiple Excel exports into one Google Sheet, applies pricing logic, and tracks unmatched keys for reconciliation, with a custom Sheets menu and dynamic file-ID handling.

Game Developer (Independent)

January 2024 – March 2024

Mega Tic-Tac-Toe

- Designed and programmed a Unity game built on a 3×3 grid of linked Tic-Tac-Toe boards where each move determines the opponent's next board, and winning a board claims one square on the grid.
- Supports local and online multiplayer, and single player vs. a Minimax-based AI with multiple difficulty levels.

Design Team Member

October 2023 – March 2024

QMIND – Queen's AI Hub

- Worked on a team project to train an agent to play Blackjack using reinforcement learning.
- Built a tabular Q-learning baseline with training loop and basic tests, then helped migrate the system to a Deep Q-learning version so the model could learn policies beyond a simple lookup table.

PERSONAL PROJECTS

trader (Python · IBKR · Finnhub · pytest)

- 24/7 AI swing-trading agent for S&P 500 equities, driven by six scheduled, idempotent routines.

Odds Aggregator (Python · Playwright · SQLite/Alembic · FastAPI)

- Production arbitrage daemon covering 10 bookmakers across 6 sports.

tax-rebalance (Python · async SQLAlchemy 2 · httpx · pytest)

- Canadian TFSA + RRSP portfolio drift monitor — emails weekly digests with cost-aware rebalance verdicts (read-only; never places trades). Spec-driven TDD: 21-task plan, two-stage independent review, 117 tests, ruff-clean, CI on Py 3.11 + 3.13. Questrade IQ API with OAuth refresh-token rotation.

equity-arbs (Python · asyncio · SQLAlchemy 2 · statsmodels · pydantic)

- Stat-arb research toolkit for TSX pair trading (spec-first, TDD, 76 unit tests). Decision-grade backtest demonstrated a ~0.6%/yr ceiling on retail capital; retired at v1.0 with the toolkit as a reusable research asset and a documented negative result.

Writing: jakethehoffer.github.io/website/#writing — short essays on engineering tradeoffs.